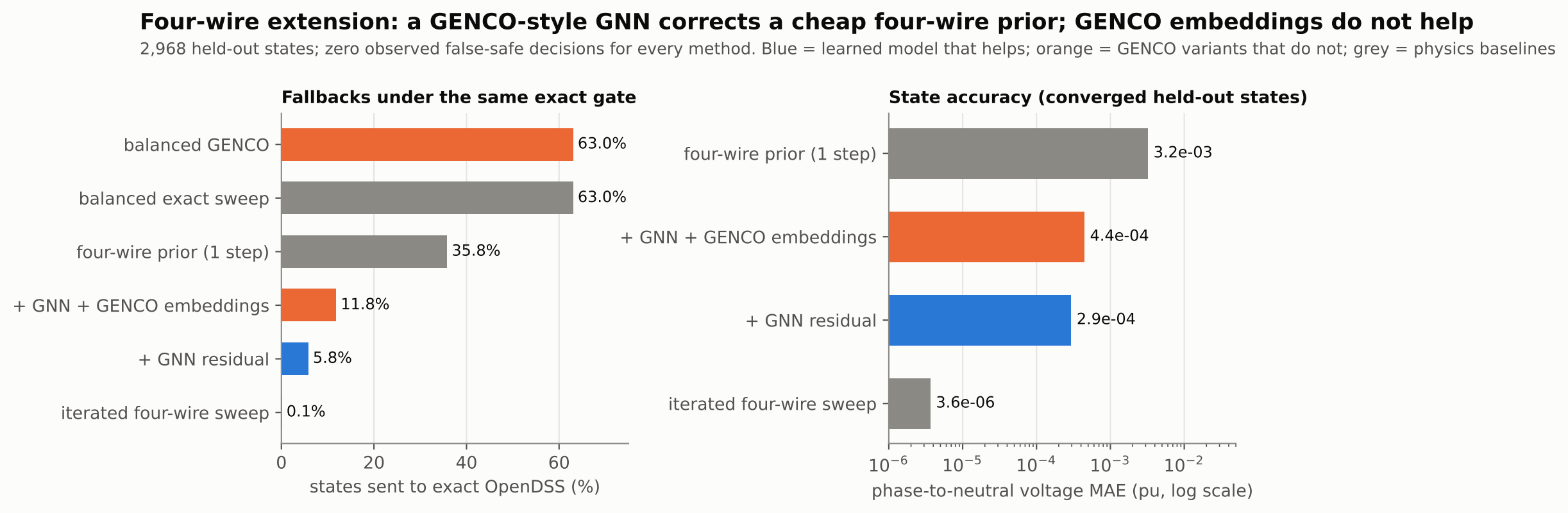
Four-wire extension: a GENCO-style GNN corrects a cheap four-wire prior; GENCO embeddings do not help
2,968 held-out states; zero observed false-safe decisions for every method. Blue = learned model that helps; orange = GENCO variants that do not; grey = physics baselines
Fallbacks under the same exact gate
balanced GENCO
balanced exact sweep
four-wire prior (1 step)
+ GNN + GENCO embeddings
+ GNN residual
iterated four-wire sweep
63.0%
63.0%
35.8%
11.8%
5.8%
0.1%
0
20
40
60
states sent to exact OpenDSS (%)
State accuracy (converged held-out states)
four-wire prior (1 step)
+ GNN + GENCO embeddings
+ GNN residual
iterated four-wire sweep
3.2e-03
4.4e-04
2.9e-04
3.6e-06
10^{-6}
10^{-5}
10^{-4}
10^{-3}
10^{-2}
phase-to-neutral voltage MAE (pu, log scale)